Riaz Hussain, PhD

Research Scientist
Center for Lung Analytics and Imaging Research (CLAIR)
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University of Alabama at Birmingham
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EDUCATION

Ph.D. Department of Mathematical, Physical and Computer Sciences, University of Parma, Parma, Italy, 2015–2019
Thesis: Spin Spectroscopy of Materials for Quantum Information Processing, Data Storage and Energy Conversion
with Doctor Europaeus Certificate

M.Phil. Department of Physics & Applied Mathematics, Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad, Pakistan, 2010–2012
Thesis: Monte Carlo-based Geant4 simulation for the gamma radiation spectrometry of compound semiconductor detectors

M.Sc. Department of Physics, University of Gujrat, Gujrat, Pakistan, 2007–2009
Major(s): Physics

B.Sc. University of Gujrat, Gujrat, Pakistan, 2005–2007
Major(s): Mathematics and Physics

PUBLICATIONS

Peer-reviewed Journal Articles

(*1*) Masokano, B. I., **Hussain**, **R.**, Garrison, W. J., Borden, A., Costa, M. L., Willmering, M. M., Kramer, E. L., Cleveland, Z. I., and Walkup, L. L. (2025). Does the Menstrual Cycle Influence Xenon Gas-Exchange MRI Metrics in Healthy Women? *ERJ Open Research*, **Accepted**.

(*2*) Casali, N., Brusaferri, A., Baselli, G., Fumagalli, S., Micotti, E., Forloni, G., **Hussain**, **R.**, Rizzo, G., and Mastropietro, A. (2026). A Comprehensive Framework for Uncertainty Quantification of Voxel-Wise Supervised Deep Learning Models in IVIM MRI. *NMR in Biomedicine 39*, e70227.

(*3*) **Hussain**, **R.**, Bdaiwi, A. S., Plummer, J. W., Masokano, B. I., Willmering, M. M., Walkup, L. L., and Cleveland, Z. I. (2026). Comparing Mean-anchored Generalized Linear Binning with Established Methods to Quantify Xenon-129 Ventilation Defect Percentage. *Academic Radiology 33*, 569–585.

(*4*) **Hussain**, **R.**, Masokano, B. I., Plummer, J. W., Bdaiwi, A. S., Willmering, M. M., Kramer, E. L., Walkup, L. L., and Cleveland, Z. I. (2025). Xe MRI Detects Ongoing Ventilation Improvements in People with Cystic Fibrosis Receiving Highly Effective Modulator Therapy. *Annals of the American Thoracic Society*, **Accepted**, DOI: `https://doi.org/10.1513/AnnalsATS.202505-497RL`.

(*5*) **Hussain**, **R.**, Plummer, J. W., Bdaiwi, A. S., Willmering, M. M., Kramer, E. L., Walkup, L. L., and Cleveland, Z. I. (2025). Optimizing Xenon-129 Ventilation MRI in Cystic Fibrosis with Spiral Imaging and Flip-angle Correction. *Radiology: Cardiothoracic Imaging 7*, PMID: 41037010, e240574.

(6) Bdaiwi, A. S., Willmering, M. M., Plummer, J. W., **Hussain**, **R.**, Walkup, L. L., Woods, J. C., and Cleveland, Z. I. (2025). Evaluating Stack-of-Stars and FLORET 3D Ultrashort Echo Time MRI to Assess Structural Pathology in Cystic Fibrosis Lung Disease. *Academic Radiology*, **Accepted**, DOI: https://doi.org/10.1016/j.acra.2025.09.036.

(7) Bdaiwi, A. S., Willmering, M. M., **Hussain**, **R.**, Woods, J. C., Walkup, L. L., and Cleveland, Z. I. (2025). Comparative evaluation of supervised and unsupervised deep learning strategies for denoising hyperpolarized ^{129}Xe lung MRI. *Magnetic Resonance in Medicine*, **Accepted**, 1–19.

(8) Bdaiwi, A. S., Willmering, M. M., Plummer, J. W., **Hussain**, **R.**, Roach, D. J., Parra-Robles, J., Niedbalski, P. J., Woods, J. C., Walkup, L. L., and Cleveland, Z. I. (2025). ^{129}Xe Image Processing Pipeline: An open-source, graphical user interface application for the analysis of hyperpolarized ^{129}Xe MRI. *Magnetic Resonance in Medicine 93*, 1220–1237.

(9) Plummer, J. W., **Hussain**, **R.**, Bdaiwi, A. S., Soderlund, S. A., Hoyos, X., Lanier, J. M., Garrison, W. J., Parra-Robles, J., Willmering, M. M., Niedbalski, P. J., Cleveland, Z. I., and Walkup, L. L. (2024). A Decay-Modeled Compressed Sensing Reconstruction Approach for non-Cartesian Hyperpolarized ^{129}Xe MRI. *Magnetic Resonance in Medicine 92*, 1363–1375.

(10) Plummer, J. W., **Hussain**, R., Bdaiwi, A. S., Costa, M. L., Willmering, M. M., Parra-Robles, J., Cleveland, Z. I., and Walkup, L. L. (2024). Analytical corrections for B_1-inhomogeneity and signal decay in multi-slice 2D spiral hyperpolarized ^{129}Xe MRI using keyhole reconstruction. *Magnetic Resonance in Medicine 92*, 967–981.

(11) Moro, F., Lisi, I., Tolomeo, D., Vegliante, G., Pascente, R., Mazzone, E., **Hussain**, **R.**, Micotti, E., Dallmeier, J., Pischiutta, F., Bianchi, E., Chiesa, R., Wang, K. K., and Zanier, E. R. (2023). Acute Blood Levels of Neurofilament Light Indicate One-Year White Matter Pathology and Functional Impairment in Repetitive Mild Traumatic Brain Injured Mice. *Journal of Neurotrauma 40*, 1144–1163.

(12) **Hussain**, **R.**, Prando, G., Selter, S., Aswartham, S., Büchner, B., and Carretta, P. (2021). Magnetically induced local lattice anomalies and low-frequency fluctuations in the Mott insulator $La_2O_3Fe_2Se_2$. *Physical Review B 103*, DOI: 10.1103/physrevb.103.1081105.

(13) Ali, I., **Hussain**, **R.**, Louis, H., Bokhari, S. W., and Iqabl, M. Z. (2021). In situ reduced graphene-based aerogels embedded with gold nanoparticles for real-time humidity sensing and toxic dyes elimination. *Microchimica Acta 188*, DOI: 10.1007/s00604-020-04658-0.

(14) Chiesa, A. et al. (2020). Understanding Magnetic Relaxation in Single-ion Magnets with high blocking temperature. *Physical Review B 101*, DOI: 10.1103/physrevb.101.174402.

(15) **Hussain**, **R.**, Cugini, F., Baldini, S., Porcari, G., Amadè, N. S., Miao, X., van Dijk, N. H., Brück, E., Solzi, M., Renzi, R. D., and Allodi, G. (2019). Ubiquitous first-order transitions and site-selective vanishing of the magnetic moment in giant magnetocaloric MnFeSiP alloys detected by ^{55}Mn NMR. *Physical Review B 100*, DOI: 10.1103/physrevb.100.104439.

(16) **Hussain**, **R.**, Allodi, G., Chiesa, A., Garlatti, E., Mitcov, D., Konstantatos, A., Pedersen, K. S., Renzi, R. D., Piligkos, S., and Carretta, S. (2018). Coherent Manipulation of a Molecular Ln-Based Nuclear Qudit Coupled to an Electron Qubit. *Journal of the American Chemical Society 140*, 9814–9818.

(17) Giansiracusa, M. J., Moreno-Pineda, E., **Hussain**, **R.**, Marx, R., Prada, M. M., Neugebauer, P., Al-Badran, S., Collison, D., Tuna, F., van Slageren, J., Carretta, S., Guidi, T., McInnes, E. J. L., Winpenny, R. E. P., and Chilton, N. F. (2018). Measurement of Magnetic Exchange in Asymmetric Lanthanide Dimetallics: Toward a Transferable Theoretical Framework. *Journal of the American Chemical Society 140*, 2504–2513.

(*18*) Farooq, M. H., **Hussain**, **R.**, Iqbal, M. Z., Shah, M., Rana, U. A., and Khan, S. U.-D. (2016). Fabrication and Magnetic Properties of Sn-Doped ZnO Microstructures via Hydrothermal Method. *Journal of Nanoscience and Nanotechnology 16*, 898–902.

(*19*) Nazar, M. F., Badshah, A., Mahmood, A., Zafar, M. N., Janjua, M. R. S. A., Raza, M. A., and **Hussain**, **R.** (2015). Synthesis, spectroscopic characterization, and computed optical analysis of green fluorescent cyclohexenone derivatives. *Journal of Physical Organic Chemistry 29*, 152–160.

(*20*) Nabi, A., **Hussain**, **R.**, Farooq, M. H., and Zubair, M. (2014). Experimental and Theoretical Investigation of Ferromagnetism in Mn-Doped SnO. *Nanoscience and Nanotechnology Letters 6*, 959–964.

(*21*) Farooq, M. H., **Hussain**, **R.**, Zhang, L., Aslam, I., Tanveer, M., Shah, M., and Iqbal, M. Z. (2014). Fabrication, characterization and magnetic properties of Mn-doped SnO nanostructures via hydrothermal method. *Materials Letters 131*, 350–353.

(*22*) Iqbal, M. Z., Wang, F., **Hussain**, **R.**, Iqbal, T., Ali, I., Rafique, M. Y., and Ali, S. (2014). Synthesis and Characterization of SnO_2 Nanorods for Energy Storage Applications. *Advanced Science, Engineering and Medicine 6*, 791–796.

(*23*) **Hussain**, **R.**, Mirza, S., and Mirza, N. (2014). The study of response of wide band gap semiconductor detectors using the Geant4. *Nuclear Technology and Radiation Protection 29*, 242–248.

(*24*) Iqbal, M. Z., Wang, F., **Hussain**, **R.**, Rafique, M. Y., Ali, S., and Ali, I. (2014). Time-Dependent Growth and Optical Properties of Stannous Oxide Architectures. *Materials Focus 3*, 92–97.

Conference Abstracts (Selected)

(*1*) **Hussain**, **R.**, Plummer, J. W., Bdaiwi, A. S., Willmering, M. M., Walkup, L. L., and Cleveland, Z. I. In *Proceedings of International Society for Magnetic Resonance in Medicine*, Honolulu, HI, USA, 2025.

(*2*) **Hussain**, **R.**, Plummer, J. W., Masokano, B. I., Bdaiwi, A. S., Willmering, M. M., Kramer, E. L., Walkup, L. L., and Cleveland, Z. I. In *Proceedings of ATS International Conference*, San Francisco, CA, USA, 2025.

(*3*) **Hussain**, **R.**, Plummer, J. W., Bdaiwi, A. S., Willmering, M. M., Walkup, L. L., and Cleveland, Z. I. In *Proceedings of ATS International Conference*, San Diego, CA, USA, 2024.

(*4*) **Hussain**, **R.**, Plummer, J. W., Willmering, M. M., Bdaiwi, A. S., Walkup, L. L., and Cleveland, Z. I. In *Proceedings of ATS International Conference*, San Diego, CA, USA, 2024.

(*5*) **Hussain**, **R.**, Plummer, J. W., Bdaiwi, A. S., Willmering, M. M., Walkup, L. L., and Cleveland, Z. I. In *Proceedings of International Society for Magnetic Resonance in Medicine*, Toronto, ON, Canada, 2023.

ACADEMIC/RESEARCH EXPERIENCE

2026–Present **Research Scientist**, University of Alabama at Birmingham, Birmingham, AL 35294, USA.

2022–2026 **Research Fellow**, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, USA.

2021–2022 **Researcher**, Mario Negri Institute for Pharmacological Research, Milan 20156, Italy.

2018–2020 **Postdoc**, University of Pavia, Pavia 27100, Italy.

2012–2015 **Research Scientist (Adjunct)**, Physics Division PINSTECH, 45650 Islamabad, Pakistan.

SCHOLARSHIPS/AWARDS

2025 **Abstract Scholarship**, Respiratory Structure and Function Assembly of the *American Thoracic Society*

2017 **Research Funding**, STFC's ISIS Neutron and Muon facility, Oxford, UK.

2015–2018 **PhD Scholarship**, Italian Ministry of Education, Universities, and Research *University of Parma, Italy*.

2010–2012 **M.Phil Fellowship**, Pakistan Institute of Engineering and Applied Sciences, Islamabad, Pakistan.

TEACHING EXPERIENCE

2018 Spring **Teaching Assistant**: NMR Lab, Physics Department, University of Parma, Parma 43124, Italy

2009–2010 **Lecturer**: Applied Physics and Applied Mechanics, Rawalpindi Swedish Institute of Technology, Rawalpindi 46300, Pakistan.

CONFERENCES AND WORKSHOPS

Oral Presentations

May 16–21, 2025 "Continued Lung Ventilation Improvement in People With Cystic Fibrosis Receiving Highly Effective Modulator Therapy", American Thoracic Society (ATS) International Conference, San Francisco, CA, USA.

May 10–15, 2025 "Median-anchored Generalized Linear Binning vs Conventional Methods to Quantify ^{129}Xe Lung Ventilation MRI", Annual Meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), Honolulu, HI, USA.

March 16–18, 2025 "Lung Function Tracking in People with Cystic Fibrosis on ETI Therapy using Xenon MRI", Ohio Valley CF Consortium, Columbous, OH, USA.

4 December 2024 "Hyperpolarized Xenon Ventilation MRI in people with Cystic Fibrosis on Highly Effective Modulator Therapy", Cincinnati Children's/University of Cincinnati CF Research Seminar, Cincinnati, OH, USA.

Feb 10–12, 2020 "Magnetic ordering and spin dynamics in $La_2O_3Fe_2Se_2$: a ^{139}La NQR study", Conference on Superconductivity and Functional Oxides (SuperFOx2020), Santa Margherita - Genoa, Italy.

Jan 30–Feb 01, 2019 "Coherent manipulation of ^{173}Yb nuclear spins coupled to Yb^{3+} qubit in Yb(trensal)", AIMAGN's 6th Italian Conference on Magnetism, Messina, Italy.

July 4–6, 2018 "Magnetic exchange and crystal field excitations of asymmetric Lanthanide dimers: An INS insight", SISN XXIX Congresso Annuale, Italian Neutron Scattering Conference, Parma, Italy.

25 May 2018 "Yb(trensal) as a Nuclear Qudit Coupled to an Electron Qubit", Molspin COST WG3 Workshop on Quantum Spin Science & Technologies, Puerto de Santiago, Tenerife, Spain.

June 3–5, 2014 "Stannous Oxide Architectures for Energy Storage Applications: A Hydrothermal Route", Conference on Frontiers of Nanoscience and Nanotechnology, Physics Division, PINSTECH, Islamabad, Pakistan.

Posters Presentations

May 17–22 2024 "Addressing Healthy Aging Variations in Ventilation from Childhood to Older Age Using Xenon MRI", American Thoracic Society (ATS) International Conference, San Diego, USA.

May 17–22 2024 "Improved Quantification of Ventilation Heterogeneity in Hyperpolarized Xenon MRI using Physics-rooted Bias-field Correction", American Thoracic Society (ATS) International Conference, San Diego, USA.

June 3–8 2023 "Improved efficiency and quantitative accuracy in hyperpolarized ^{129}Xe ventilation imaging using 2D spiral acquisition", Annual Meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), Toronto, Canada.

May 5 2023 "Physics works: Improved functional lung MR images with perks", First Annual CCHMC Research Symposium, Cincinnati OH, USA.

Mar 1–9, 2018 "Let's view it from the nucleus: Yb Single Ion Magnet for QIP", 38th Berlin School on Neutron Scattering, HZB Berlin, Germany.

TECHNICAL SKILLS

Hands-on Experience

MRI (Clinical and Preclinical), NMR/NQR, MuSR, Neutron Scattering.

Programming & Analysis Tools

Python, Matlab, R, LaTeX, FSL, ANTs, ITKSnap, SPM, Philips Pulse Programming Courses

SOCIETY MEMBERSHIPS

Member American Thoracic Society (ATS)

Member International Society for Magnetic Resonance in Medicine

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